



BAULKHAM HILLS HIGH SCHOOL

2017
YEAR 11 TASK 2

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes.
- Working time – 1 hour.
- Write using non erasable black or blue pen.
- Board-approved calculators may be used
- All relevant mathematical reasoning and/or calculations must be shown

Total marks – 36

This paper consists of 3 Questions.

Start each question on the appropriate page of your answer booklet

In Questions 1-3, your responses should include relevant mathematical reasoning and/ or calculations.

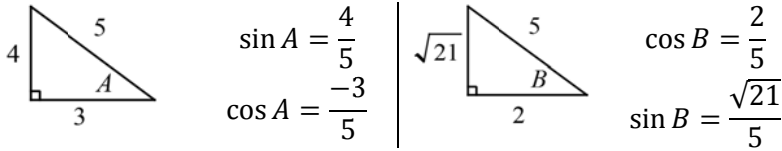
Question 1 (12 marks)		Marks
(a)	Find the exact value of $\tan 105^\circ$.	2
(b)	Solve for x if $\sin 2x = \cos x \text{ for } 0^\circ \leq x \leq 360^\circ$	3
(c)	Find the acute angle between the lines $x - y + 2 = 0$ and $2x - y - 4 = 0$. Give your answer to the nearest degree.	2
(d)	Find the point P which divides the interval AB internally in the ratio of 2:3, where A is $(-4, -9)$ and B is $(5, 9)$.	2
(e)	Simplify the following function in terms of t , if $t = \tan \frac{x}{2}$: $\frac{\sin x - \cos x + 1}{\sin x + \cos x - 1}$	3
End of Question 1		

Question 2 (12 marks)		Marks
(a)	By using $t = \tan \frac{x}{2}$, solve for x $3 \sin x - 2 \cos x = 2 \text{ for } 0 \leq x \leq 360^\circ.$ Give your answer to the nearest degree.	3
(b)	Given that the point $P(13,3)$ lies on the line joining $A(7,1)$ and $B(-5,-3)$, find the ratio in which P divides the interval AB .	2
(c)	(i) Express $\sqrt{3}\cos x + \sin x$ in the form $R \cos(x - \alpha)$ where α is in degrees.	2
	(ii) Hence, find the solution of the equation. $\sqrt{3}\cos x + \sin x = \frac{11}{6} \text{ for } 0^\circ \leq x \leq 360^\circ.$ Give your answer to the nearest degree.	3
(d)	If $\sin A = \frac{4}{5}$ where A is obtuse and $\cos B = \frac{2}{5}$ where B is acute, find the exact value of $\sin(A + B)$	3
End of Question 2		

Question 3 (11 marks)		
(a)	(i) Prove that $\sqrt{\frac{1-\cos 2x}{1+\cos 2x}} = \tan x$, where x is acute.	2
	(ii) Hence find a and b if $\tan 15^\circ = a - \sqrt{b}$.	3
(b)	(i) Prove that $\frac{\sin 3x}{\sin x} + \frac{\cos 3x}{\cos x} = 4 \cos 2x$	3
	(ii) Hence solve for $\frac{\sin 3x}{\sin x} + \frac{\cos 3x}{\cos x} = 4 \cos^2 2x$ for $0^\circ \leq x \leq 180^\circ$	3
End of Exam		

Year 11 Task 2 X1 2017

	Solutions	Mks	Comments
1a	$\begin{aligned}\tan 105^\circ &= \tan(60^\circ + 45^\circ) \\ &= \frac{\tan 60^\circ + \tan 45^\circ}{1 - \tan 60^\circ \tan 45^\circ} \\ &= \frac{\sqrt{3} + 1}{1 - \sqrt{3}} \\ &= -2 - \sqrt{3}\end{aligned}$	2	2 marks • correct solution. 1 mark • Correct expansion • Finds exact values
1b	$\begin{aligned}\sin 2x &= \cos x \\ 2 \sin x \cos x - \cos x &= 0 \\ \cos x (2 \sin x - 1) &= 0\end{aligned}$ $\begin{array}{ll}\cos x = 0 & \sin x = \frac{1}{2} \\ x = 90^\circ \text{ or } 270^\circ & x = 30^\circ \text{ or } 150^\circ\end{array}$ $\therefore x = 30^\circ, 90^\circ, 150^\circ, 270^\circ$	3	3 marks • correct solution. 2 marks • Solving $\cos x = 0$ or $\sin x = \frac{1}{2}$ 1 mark • expanding $\sin 2x$.
1c	$\begin{aligned}m_1 &= 1, m_2 = 2 \\ \tan \theta &= \left \frac{2 - 1}{1 + 2 \times 1} \right \\ \tan \theta &= \frac{1}{3} \\ \theta &= 18^\circ 26' \\ \theta &= 18^\circ \text{ (nearest degree)}\end{aligned}$	2	2 marks • correct solution. 1 mark • correct gradients • correct use of formula
1d	$\begin{aligned}\left(\frac{-4 \times 3 + 2 \times 5}{2 + 3}, \frac{-9 \times 3 + 2 \times 9}{2 + 3} \right) \\ = \left(\frac{-2}{5}, \frac{-9}{5} \right)\end{aligned}$	2	2 marks • correct solution. 1 mark • correct use of formula • One ordinate correct
1e	$\begin{aligned}\frac{\sin x - \cos x + 1}{\sin x + \cos x - 1} &= \frac{\left(\frac{2t}{1+t^2} \right) - \left(\frac{1-t^2}{1+t^2} \right) + 1}{\left(\frac{2t}{1+t^2} \right) + \left(\frac{1-t^2}{1+t^2} \right) - 1} \\ &= \frac{2t - 1 + t^2 + 1 + t^2}{2t + 1 - t^2 - 1 - t^2} \\ &= \frac{2t^2 + 2t}{2t - 2t^2} \\ &= \frac{2t(t+1)}{2t(1-t)} \\ &= \frac{t+1}{1-t}\end{aligned}$	3	3 marks • correct solution. 2 marks • significant progress towards solution 1 mark • correct substitution of t

2a	$3\left(\frac{2t}{1+t^2}\right) - 2\left(\frac{1-t^2}{1+t^2}\right) = 2$ $6t - 2 + 2t^2 = 2 + 2t^2$ $\tan \frac{x}{2} = \frac{2}{3}$ $\frac{x}{2} = 33.69^\circ$ $x = 67^\circ$ Test $x = 180^\circ$ $LHS = 3 \sin 180^\circ - 2 \cos 180^\circ$ $= 2$ $= RHS$ $\therefore x = 67^\circ \text{ or } 180^\circ$	3	3 marks • correct solution. 2 marks • Finds $x = 67^\circ$ using t -method 1 mark • expanding $\sin 2x$. 0 mark • solving without using t method.
2b	$x_{AP} = 13 - 7 = 6$ $x_{PB} = 13 - (-5) = 18$ $x_{AP} : x_{PB} = 6 : -18$ $\therefore \text{ratio } AP : PB = 1 : -3$	2	2 marks • correct solution. 1 mark • correct lengths
2c(i)	$\sqrt{3}\cos x + \sin x = R \cos x \cos \alpha + R \sin x \sin \alpha$ $R \cos \alpha = \sqrt{3}$ $R \sin \alpha = 1$ $\tan \alpha = \frac{1}{\sqrt{3}}$ $\alpha = 30^\circ$ $R = 2$ $\sqrt{3}\cos x + \sin x = 2 \cos(x - 30^\circ)$	2	2 marks • correct solution. 1 mark • correct R • correct α
2c(ii)	$2 \cos(x - 30^\circ) = \frac{11}{6}$ $\cos(x - 30^\circ) = \frac{11}{12}$ $x - 30^\circ = \pm 23.55^\circ \text{ for } -30^\circ \leq x \leq 330^\circ$ $x = 6^\circ \text{ or } 54^\circ$	3	3 marks • correct solution. 2 marks • Find the 2 solutions for $(x - 30^\circ)$ 1 mark • Finds acute angle of 23.55°
2d	 $\sin A = \frac{4}{5}$ $\cos A = \frac{-3}{5}$ $\cos B = \frac{2}{5}$ $\sin B = \frac{\sqrt{21}}{5}$ $\sin(A + B) = \sin A \cos B + \sin B \cos A$ $\sin(A + B) = \frac{4}{5} \times \frac{2}{5} + \frac{\sqrt{21}}{5} \times -\frac{3}{5}$ $= \frac{8 - 3\sqrt{21}}{25}$	3	3 marks • correct solution. 2 marks • correct expansion and use of triangles to find trig ratios 1 mark • correct expansion of $\sin(A + B)$ • Uses triangles to find exact trig ratios.

3a(i)	$LHS = \sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}}$ $= \sqrt{\frac{1 - (2 \cos^2 x - 1)}{1 + (2 \cos^2 x - 1)}}$ $= \sqrt{\frac{2(1 - \cos^2 x)}{2 \cos^2 x}}$ $= \sqrt{\frac{\sin^2 x}{\cos^2 x}}$ $= \tan x$	2	2 marks • correct solution. 1 mark • correct use of $\cos 2x$ identity
3a(ii)	$\tan 15^\circ = \sqrt{\frac{1 - \cos 2 \times 15^\circ}{1 + \cos 2 \times 15^\circ}}$ $\tan 15^\circ = \sqrt{\frac{1 - \frac{\sqrt{3}}{2}}{1 + \frac{\sqrt{3}}{2}}}$ $= \sqrt{\frac{2 - \sqrt{3}}{2 + \sqrt{3}}}$ $= \sqrt{\frac{(2 - \sqrt{3})^2}{4 - 3}}$ $= 2 - \sqrt{3}$ $\therefore a = 2, b = 3$	3	3 marks • correct solution. 2 marks • $\sqrt{7 - 4\sqrt{3}}$ • substitutes values and rationalises 1 mark • correct substitution for $x = 15^\circ$ (Maximum of 1 mark if not using hence)
3b(i)	$LHS = \frac{\sin 3x}{\sin x} + \frac{\cos 3x}{\cos x}$ $= \frac{\sin 3x \cos x + \cos 3x \sin x}{\sin x \cos x}$ $= \frac{\sin(3x + x)}{\sin x \cos x}$ $= \frac{\sin 4x}{\sin x \cos x}$ $= \frac{2 \sin 2x \cos 2x}{\sin x \cos x}$ $= \frac{2 \times 2 \sin x \cos x \cos 2x}{\sin x \cos x}$ $= 4 \cos 2x$	3	3 marks • correct solution. 2 marks • significant progress towards solution 1 mark • identifies $\sin(5x - x)$

3b(ii)	$4 \cos 2x = 4 \cos^2 2x$ $\cos 2x - \cos^2 2x = 0$ $\cos 2x (1 - \cos 2x) = 0$ <table><tr><td>$\cos 2x = 0$ $2x = 90^\circ, 270^\circ$ $x = 45^\circ, 135^\circ$</td><td>$\cos 2x = 1$ $2x = 0^\circ, 360^\circ$ $x = 0^\circ, 180^\circ$</td></tr></table> But $\sin x \neq 0$ and $\cos x \neq 0$ <u>$\therefore x = 45^\circ, 135^\circ.$</u>	$\cos 2x = 0$ $2x = 90^\circ, 270^\circ$ $x = 45^\circ, 135^\circ$	$\cos 2x = 1$ $2x = 0^\circ, 360^\circ$ $x = 0^\circ, 180^\circ$	3	3 marks • correct solution. 2 marks • Solves for $\cos 2x = 0$ and $\cos 2x = 1$ 1 mark • uses equation in (i) to from a simplified equivalent expression.
$\cos 2x = 0$ $2x = 90^\circ, 270^\circ$ $x = 45^\circ, 135^\circ$	$\cos 2x = 1$ $2x = 0^\circ, 360^\circ$ $x = 0^\circ, 180^\circ$				